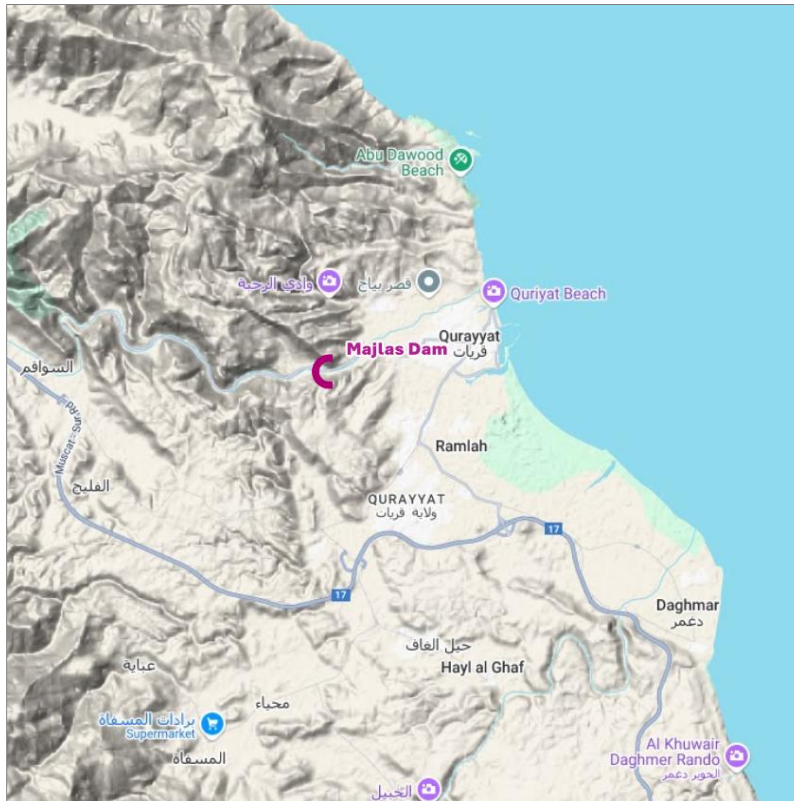




Sultanate of Oman
Ministry of Agriculture, Fisheries & Water Resources

Consultancy Services for Design Review and Supervision of Wadi Majlas Dam in Wilayat Qurayat, Muscat Governorate



Wadi Majlas Flood Protection Dam Hydrology Report

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 **RENARDET** S.A. & Partners
CONSULTING ENGINEERS L.L.C.



TABLE OF CONTENTS

Executive Summary	1
1. Introduction	2
2. Study Area and Physiography	3
2.1. Meteorology	4
3. Available Data.....	6
4. Statistical Analysis of Rainfall Data	7
4.1. Methodology	7
4.2. At-site results	7
4.3. Regional growth curve and representative depths	8
4.4. Intensity-Duration-Frequency (IDF) relationships	9
5. Comparison with the Oman Flood-Mapping Rainfall	11
6. Probable Maximum Precipitation (PMP)	13
7. Rainfall-Runoff Model	15
7.1. SCS curve-number loss method.....	15
7.2. Curve Number grid.....	15
7.3. Area Reduction Factor	16
7.4. Catchment and stream delineation	17
7.5. Sub-basin characteristics and lag time	18
8. Model Calibration and Review of the Previous Study	20
9. Design Flood Results	21
9.1. Peak discharges	21
9.2. Reservoir routing and attenuation.....	22
10. Conclusions and Recommendations	25
11. References.....	26



LIST OF FIGURES

Wadi Majlas catchment - topography, drainage network and rain-gauge locations (UTM 40N).....	3
Monthly air temperature at Nakhal station (representative of the Majlas area).	4
Wind rose at the Abyad reference point (ERA-5 wind components).	5
24-hour regional dimensionless growth curve (pooled gauges) with at-site data.	8
Representative Intensity-Duration-Frequency (IDF) curves for the Wadi Majlas study area (GEV/L-moments).	10
24-hour design rainfall - local RFA versus Oman Flood-Mapping for the Majlas study area.	12
24-hour Probable Maximum Precipitation over the Wadi Majlas area (WGS-84).....	14
SCS curve-number grid (GCN250) over the Wadi Majlas catchment.	16
HEC-HMS sub-basin delineation of the Wadi Majlas model.....	18
Majlas Dam elevation-storage-area (hypsothetic) curves.	21
Peak-discharge frequency at the catchment outlet - effect of the Majlas reservoir.	22
Majlas reservoir routing - peak inflow versus routed outflow by return period.....	23
Majlas reservoir inflow and routed-outflow hydrographs (10,000-year and PMP events).	24



LIST OF TABLES

Physiographic characteristics of the Wadi Majlas catchment (from DTM_5m).....	3
At-site 24-hour design rainfall depths (mm) by GEV/L-moments.....	7
Representative 24-hour design rainfall depths for the study area (local GEV/L-moment RFA).	9
24-hour design rainfall: local RFA versus Oman Flood-Mapping.	11
Area reduction factor applied in the depth-area analysis.	16
Characteristics of the HEC-HMS sub-basins (Wadi Majlas model).	19
Rainfall reduction required to reproduce the questioned observed hydrograph (Cyclone Gonu, 2007). ...	20
Design peak discharges (m ³ /s) at key locations, with and without the Majlas reservoir.	22
Majlas reservoir routing - peak inflow, routed outflow and attenuation.	23



EXECUTIVE SUMMARY

This report presents the updated hydrological study for the **Wadi Majlas Flood Protection Dam**, Sultanate of Oman. It supersedes the hydrology chapter of the previous design report and documents the rainfall analysis, Probable Maximum Precipitation (PMP), rainfall-runoff modelling, calibration review and design flood estimates that underpin the dam and the downstream dyked-channel protection scheme.

The Wadi Majlas catchment draining to the proposed dam has an area of about **571 km²**, rising from the dam site to over 1,950 m in the Hajar mountains. Including the additional downstream sub-catchments introduced for the dyked-channel design, the modelled system covers **604 km²**. A HEC-HMS model was built on a 5 m NSA digital terrain model, an SCS curve-number loss method (GCN250 grid, catchment-average CN of about 89) and Kirpich lag times, and driven by 24-hour design storms with a depth-area reduction factor.

Two rainfall bases were examined. A local regional-frequency analysis (RFA) of nine rain gauges, fitted with the **Generalised Extreme Value (GEV) distribution using L-moments**, was compared with the Oman Flood-Mapping design rainfall that was adopted to drive the HEC-HMS runs. The two agree closely at short return periods (2-year: 27.9 mm versus 27.6 mm) but the local analysis produces a much heavier tail at rare return periods, because the short local records are dominated by the exceptional 2007 Cyclone Gonu event. The Flood-Mapping depths were therefore retained as the design basis.

A review of the previous calibration found that one of the three observed Gonu hydrographs used near the Majlas dam site could only be reproduced by reducing gauged rainfall to physically implausible levels. That observed hydrograph is judged to be erroneous and should not have been used for calibration; this was not flagged in the previous report.

Design peak discharges were derived for return periods from 2 to 10,000 years plus the PMP, with and without the attenuating effect of the Majlas reservoir. The reservoir stores the full inflow up to about the 200-year event and substantially attenuates larger floods: for the 10,000-year event the peak at the dam is reduced from about 14,170 m³/s (inflow) to 8,750 m³/s (routed outflow). These results provide the hydrological design inputs for the dam spillway and the downstream dyked channel.



1. INTRODUCTION

The Ministry of Agriculture, Fisheries and Water Resources is developing a flood-protection scheme for Wadi Majlas comprising a flood-detention dam and a downstream dyked (leveed) channel. This report updates the hydrological basis of the scheme. The objectives of the study are to:

- characterise the Wadi Majlas catchment and its hydro-meteorological setting;
- derive design rainfall for durations up to 24 hours and return periods up to 10,000 years, and estimate the PMP;
- build and calibrate a HEC-HMS rainfall-runoff model of the catchment and the downstream reaches;
- review the calibration adopted in the previous study against the observed cyclone hydrographs; and
- produce design flood hydrographs and peak discharges for the dam and the downstream protection works.

The study builds on, and where necessary corrects, the earlier hydrology report. Two HEC-HMS models were developed: an initial model used to construct and calibrate the catchment response against observed hydrographs and peak flows, and a final model that adds the downstream sub-catchments required for the dyked-channel design and applies a depth-area analysis to represent the areal reduction of extreme rainfall.



2. STUDY AREA AND PHYSIOGRAPHY

The Wadi Majlas catchment drains the eastern Hajar mountains towards the dam site. The contributing area to the dam is **571 km²**. Elevations range from about 80 m a.s.l. at the dam to nearly 1,970 m at the highest divide, giving a strongly mountainous, high-relief catchment with a dense, steep drainage network. Physiographic parameters derived from the 5 m NSA digital terrain model (DTM_5m) are summarised in Table 1.

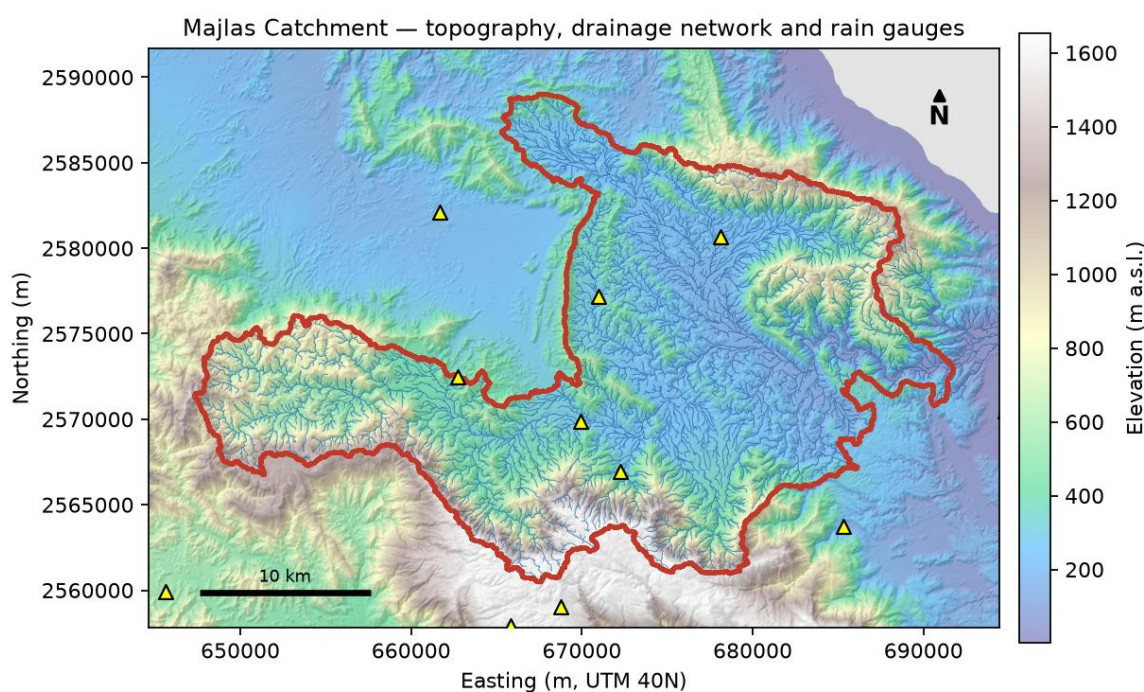


Figure 1 Wadi Majlas catchment - topography, drainage network and rain-gauge locations (UTM 40N).

Table 1 Physiographic characteristics of the Wadi Majlas catchment (from DTM_5m).

Physiographic parameter	Value
Contributing area to dam	571 km ²

Physiographic parameter	Value
Catchment-average curve number (CN)	89
Time of concentration, T _c	400 min
Longest flow path length	68.4 km
Longest flow-path slope	0.0216 m/m
10-85% flow-path length	51.3 km
10-85% flow-path slope	0.0088 m/m
Mean basin slope	0.546 m/m
Basin relief	1,948 m
Elongation ratio	0.394
Drainage density	0.161 km/km ²

2.1. METEOROLOGY

The regional climate is hot arid, with intense but infrequent rainfall associated with tropical cyclones and convective storms. Monthly air-temperature statistics for the representative Nakhal station are shown in Figure 2; mean monthly temperatures range from about 24 C in winter to 35 C in summer, with recorded maxima above 50 C. Prevailing winds derived from ERA-5 components at the Abyad reference point are summarised in the wind rose of Figure 3. These parameters support the moisture-maximisation used in the PMP estimate and the reservoir evaporation assessment.

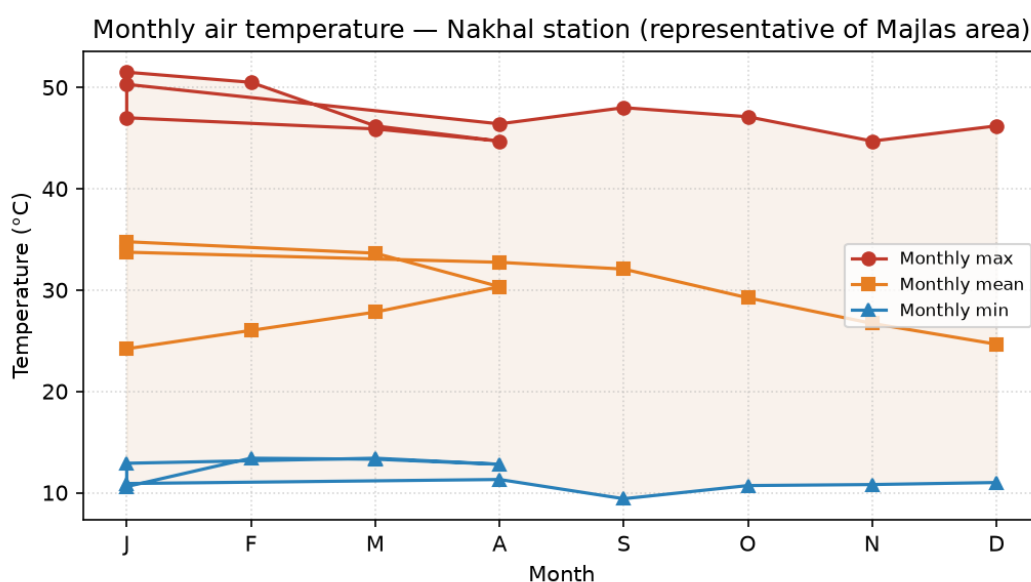


Figure 2 Monthly air temperature at Nakhal station (representative of the Majlas area).



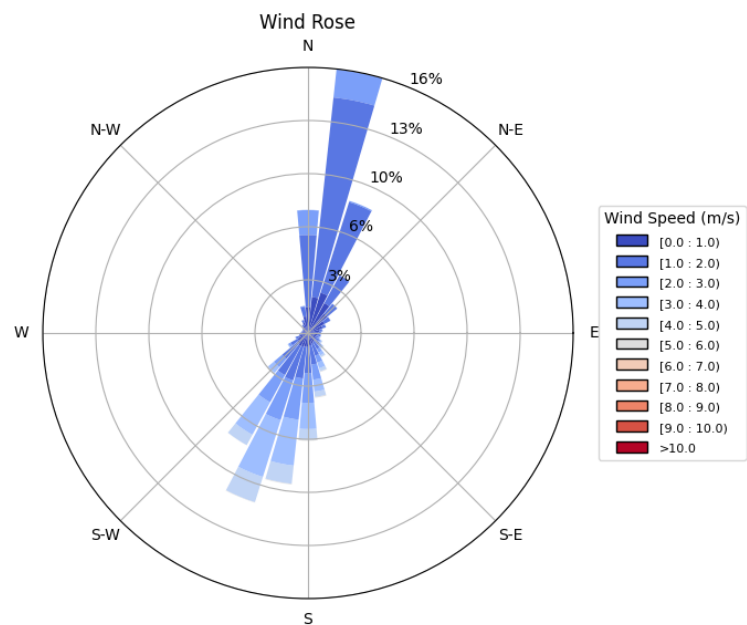


Figure 3 Wind rose at the Abyad reference point (ERA-5 wind components).

3. AVAILABLE DATA

The hydrological analysis draws on the following data sets:

- Sub-daily rainfall records from nine gauges in and around the catchment (1994-2024), supplied as event-based intensity series;
- The Oman Flood-Mapping regional design rainfall (24-hour depths and 10-minute temporal distributions for return periods 2-PMP);
- A 5 m NSA digital terrain model (DTM_5m) for catchment and stream delineation and physiography;
- The GCN250 global curve-number grid, clipped to the catchment;
- Reservoir elevation-area-storage (hypsothetic) data from the drone survey of the dam site; and
- Observed cyclone hydrographs (Gonu 2007, Phet 2010) used for calibration in the previous study.



4. STATISTICAL ANALYSIS OF RAINFALL DATA

4.1. METHODOLOGY

To improve the reliability of rainfall-frequency estimation within the Wadi Majlas catchment, a regional index-flood (pooled) frequency analysis was adopted rather than independent single-site analyses. Annual maximum series (AMS) from the gauges influenced by homogeneous synoptic and orographic conditions are combined, after normalisation by their at-site mean, into a single dimensionless growth curve. Pooling increases the effective record length and reduces sampling uncertainty for rare events.

The pooled data were fitted with the **Generalised Extreme Value (GEV) distribution, with parameters estimated by the method of L-moments** (Hosking & Wallis, 1997). Only the GEV/L-moment estimator is used in this study. This choice is deliberate and is justified as follows:

- L-moments are linear combinations of order statistics and are far less sensitive to the few very large cyclone outliers that dominate short arid-zone records than conventional product moments;
- the GEV distribution is the asymptotic form for block (annual) maxima and is the international standard for regional rainfall frequency analysis;
- using a single, well-understood estimator keeps the analysis transparent and reproducible and avoids the inconsistency of mixing distributions or fitting methods across sites and durations.

Annual maxima were extracted directly from the observed records with no gap-filling. Each gauge series was placed on a fixed 10-minute grid; intervals with no record were treated as dry (zero) and genuine recorded zeros were retained as zeros. For every calendar year containing data, the annual maximum rolling depth was computed for each duration. Avoiding infilling removes the risk of propagating synthetic values into later gaps (i.e. filling gaps from previously filled values) and of mis-classifying true zeros as missing data - both of which distort the extreme-value tail. The GEV/L-moment estimator is robust to the resulting variable record lengths.

4.2. AT-SITE RESULTS

The at-site 24-hour AMS were fitted with GEV/L-moments for the five gauges with records of at least eight years. The resulting design depths (selected return periods) are given in Table 2. The L-skewness of every long record is high (0.59-0.78), reflecting the dominance of the 2007 Cyclone Gonu event (500-625 mm in 24 h) over otherwise modest annual maxima (typically 10-50 mm).

Table 2 At-site 24-hour design rainfall depths (mm) by GEV/L-moments.



Gauge	n (yr)	Mean (mm)	2-yr	10-yr	100-yr	1,000-yr	10,000-yr
Jabal Abyadh East	13	108	48	208	948	3,917	15,887
Buei near Buei	20	53	23	96	469	2,127	9,523
Station FL762629	7	31	24	67	140	242	386
Taba at Taba	21	58	22	92	512	2,701	14,165
Hayfadh at Hayfadh	22	52	15	76	484	2,860	16,762
Qurayat at Qurayat	19	47	25	88	351	1,312	4,836
Station FM865337	6	46	35	94	209	394	692

4.3. REGIONAL GROWTH CURVE AND REPRESENTATIVE DEPTHS

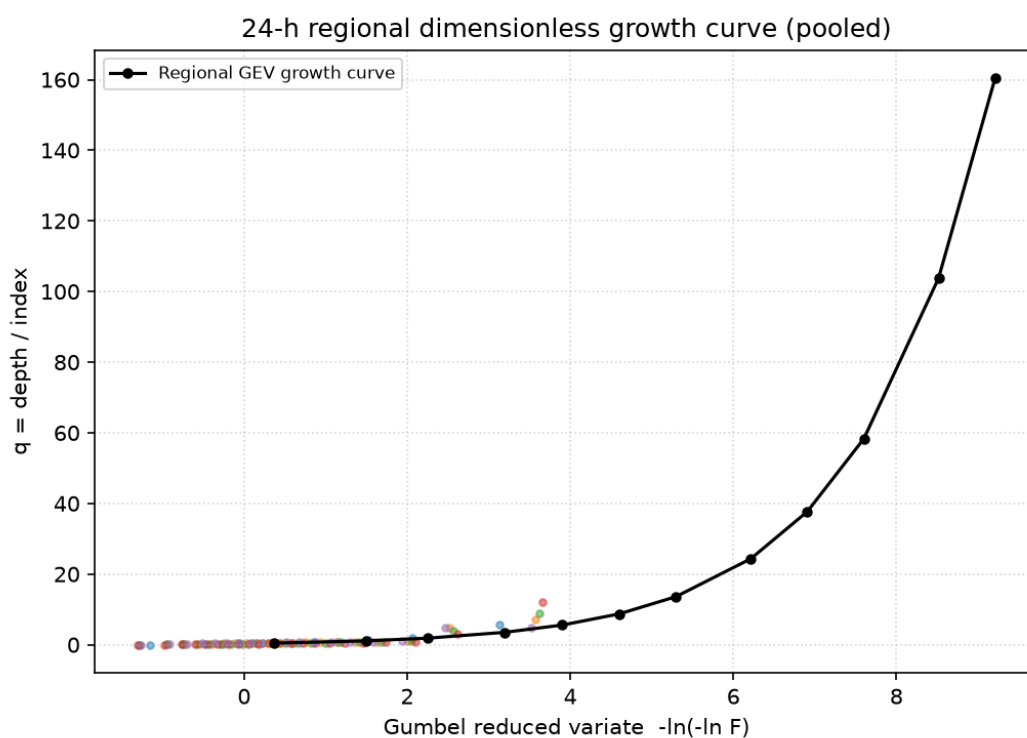


Figure 4 24-hour regional dimensionless growth curve (pooled gauges) with at-site data.

Normalising each gauge AMS by its at-site mean and pooling gives the regional growth curve of Figure 4. Multiplying the growth factors by the catchment-representative index rainfall (mean of the pooled at-site means, 63.5 mm) yields the representative 24-hour design depths for the study area listed in Table 3.



Table 3 Representative 24-hour design rainfall depths for the study area (local GEV/L-moment RFA).

Return period (yr)	24-h depth (mm)
2	28
5	69
10	118
25	223
50	354
100	554
200	864
500	1,546
1000	2,395
2000	3,706
5000	6,594
10000	10,192

4.4. INTENSITY-DURATION-FREQUENCY (IDF) RELATIONSHIPS

IDF relationships were built by applying the single regional growth curve to the index rainfall of each duration (10 min to 24 h). Because all durations share one growth curve, the return-period curves remain parallel and do not cross or diverge at durations beyond one hour - an artefact seen when each duration is fitted independently. The representative IDF curves are shown in Figure 5.



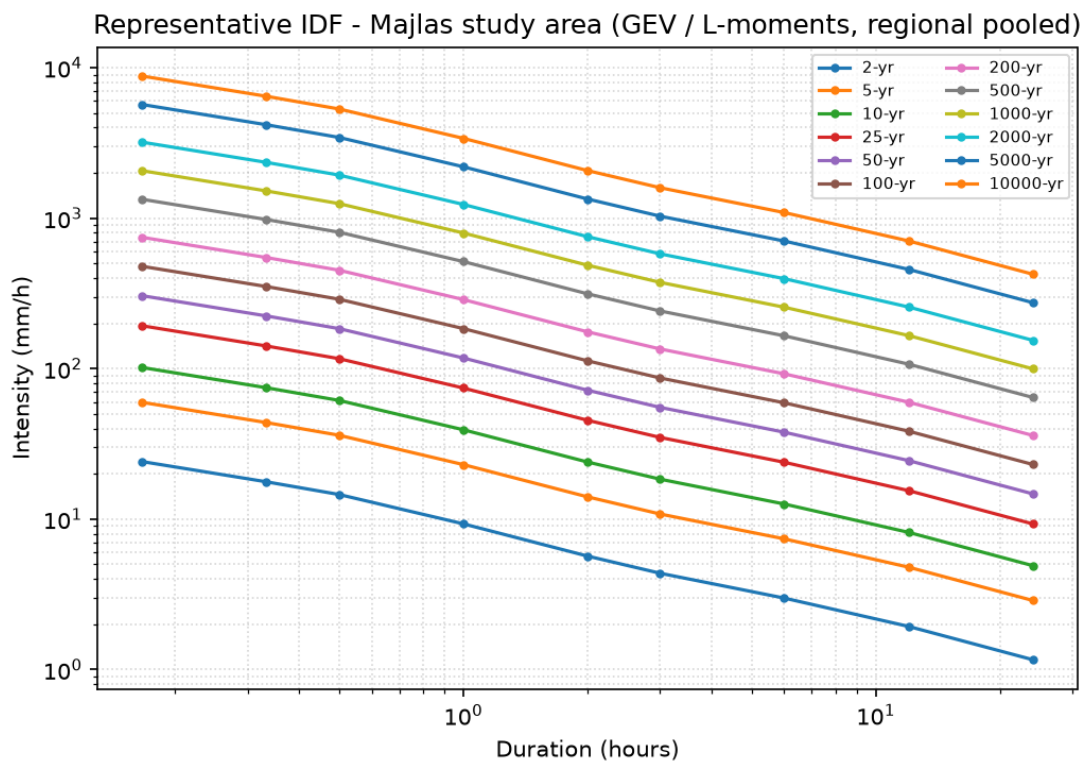


Figure 5 Representative Intensity-Duration-Frequency (IDF) curves for the Wadi Majlas study area (GEV/L-moments).

5. COMPARISON WITH THE OMAN FLOOD-MAPPING RAINFALL

The Oman Flood-Mapping study provides regional 24-hour design rainfall covering the Majlas catchment; these depths were used to drive all HEC-HMS design runs in this study. Table 4 and Figure 6 compare the local GEV/L-moment depths derived above with the Flood-Mapping depths.

Table 4 24-hour design rainfall: local RFA versus Oman Flood-Mapping.

Return period (yr)	Local GEV (mm)	Flood Mapping (mm)	Ratio
2	28	28	1.01
5	69	55	1.25
10	118	106	1.12
25	223	138	1.62
50	354	168	2.10
100	554	205	2.70
200	864	251	3.44
500	1,546	327	4.72
1000	2,395	400	5.99
2000	3,706	489	7.58
5000	6,594	639	10.32
10000	10,192	685	14.88

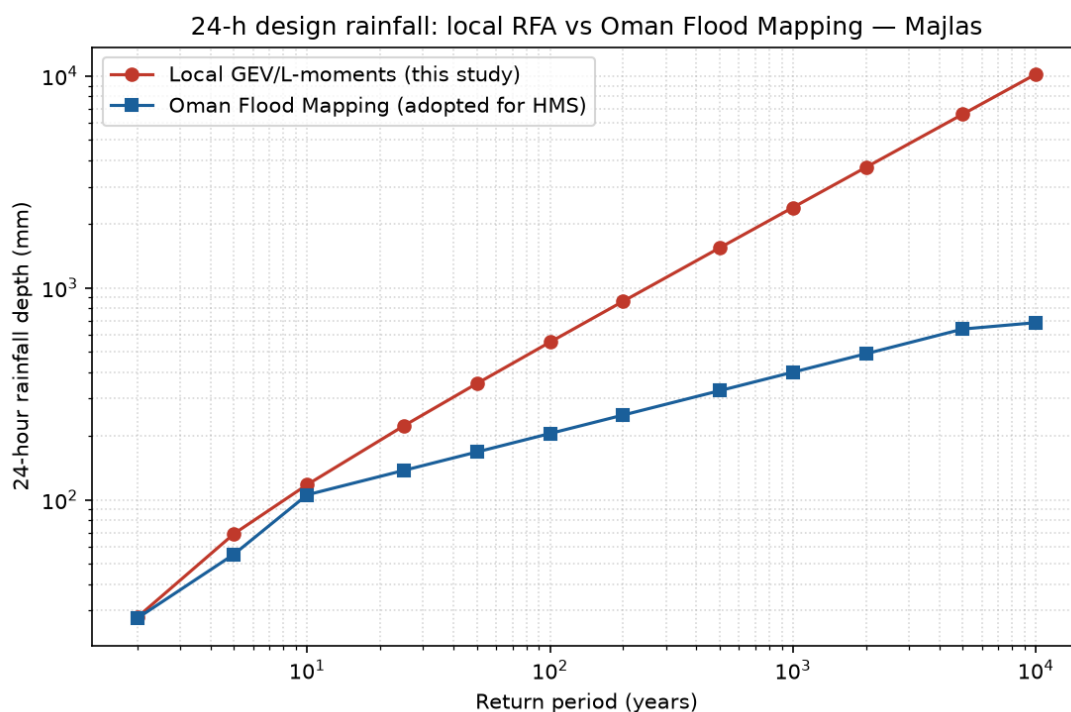


Figure 6 24-hour design rainfall - local RFA versus Oman Flood-Mapping for the Majlas study area.

The two data sets are almost identical at the 2-year event (27.9 mm versus 27.6 mm) and remain within about 25% up to the 10-year event. Beyond this, the local analysis diverges rapidly upward, reaching roughly three times the Flood-Mapping depth at 100 years and an order of magnitude at 10,000 years. This divergence is a direct consequence of the short local records being dominated by a single, exceptionally rare event (Cyclone Gonu, 2007), which produces a heavy-tailed growth curve when extrapolated far beyond the observation period. The Flood-Mapping depths, which draw on a longer and spatially extensive regional database, are more robust for rare return periods and were therefore **adopted as the design rainfall for the HEC-HMS modelling**. The local analysis is retained as an independent check that confirms agreement in the well-observed frequency range and highlights the sensitivity of the tail to the Gonu event.



6. PROBABLE MAXIMUM PRECIPITATION (PMP)

The Probable Maximum Precipitation is defined by the World Meteorological Organization (WMO, 1986) as the greatest depth of precipitation that is meteorologically possible for a given duration over a particular basin. Two approaches are recognised: a statistical (Hershfield) approach based on maximising observed series, and a deterministic moisture-maximisation and storm-transposition approach. The latter, applied in the regional Oman PMP study, was adopted here.

Historical cyclones (Gonu 2007, Phet 2010, Keila 2011, Mekunu 2018 and Shaheen 2021) were reviewed to select a reference storm. Cyclone Gonu (2007), with the highest recorded 24-hour rainfall in the region, was maximised and transposed to the project area. The storm rainfall is maximised by the ratio of precipitable water under maximum and observed dew-point conditions:

$$PMP = R_0 \frac{W_m}{W_s}$$

where R_0 is the observed storm rainfall, and W_m and W_s are the precipitable-water depths corresponding to the maximum and observed dew points respectively. ERA-5 dew-point temperatures (1940-present) were used to derive the maximising and transposition factors. The resulting 24-hour PMP grid across the project area is shown in Figure 7; over the Wadi Majlas catchment the PMP ranges from about 780 mm at the dam to over 1,600 mm on the highest ground, consistent with the design value adopted from the regional PMP grid.



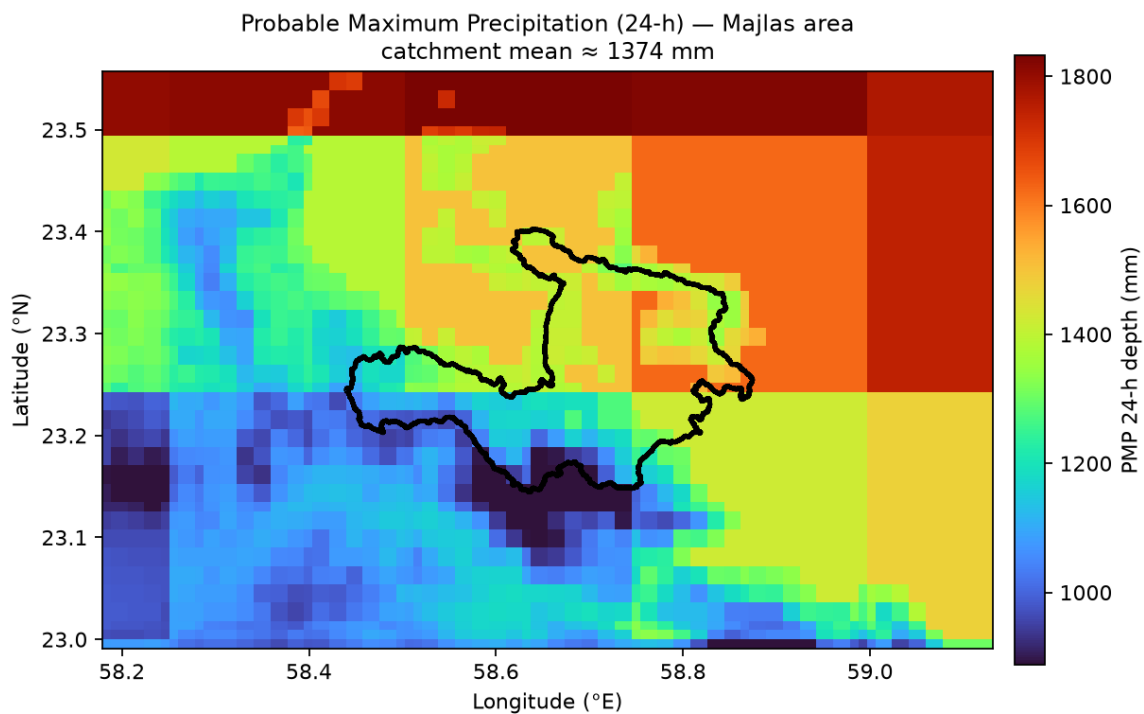


Figure 7 24-hour Probable Maximum Precipitation over the Wadi Majlas area (WGS-84).

7. RAINFALL-RUNOFF MODEL

The catchment response was simulated in HEC-HMS using the SCS curve-number loss method, the SCS unit hydrograph transform with Kirpich lag times, and reach routing between sub-basins. Design storms were applied as 24-hour hyetographs with the Flood-Mapping temporal distribution and a depth-area reduction factor.

7.1. SCS CURVE-NUMBER LOSS METHOD

Direct runoff is computed from the SCS relationship:

$$Q = \frac{(P - I_a)^2}{(P - I_a + S)}$$

where Q is runoff depth, P is rainfall depth, I_a is the initial abstraction and S is the potential maximum retention, related to the curve number CN by:

$$S = \frac{25400}{CN} - 254 \text{ (mm)}$$

$$I_a = 0.2 S$$

The variation in infiltration among soils is represented through the four hydrologic soil groups (A-D). Arid- and semi-arid rangeland curve numbers follow the Oman Hydrological Design Standards (OHDS-2017); for the mountainous terrain of the catchment the applicable values are CN 86-90 (slope < 18%) and 90-94 (slope > 18%).

7.2. CURVE NUMBER GRID

Spatially distributed curve numbers were taken from the GCN250 global data product (250 m resolution, combining ESA CCI land cover and HYSOGs250m soil groups) and clipped to the catchment (Figure 8). The resulting catchment-average curve number is approximately **89** (AMC II), with sub-basin values in the range 86.5-91.0 (Table 7).



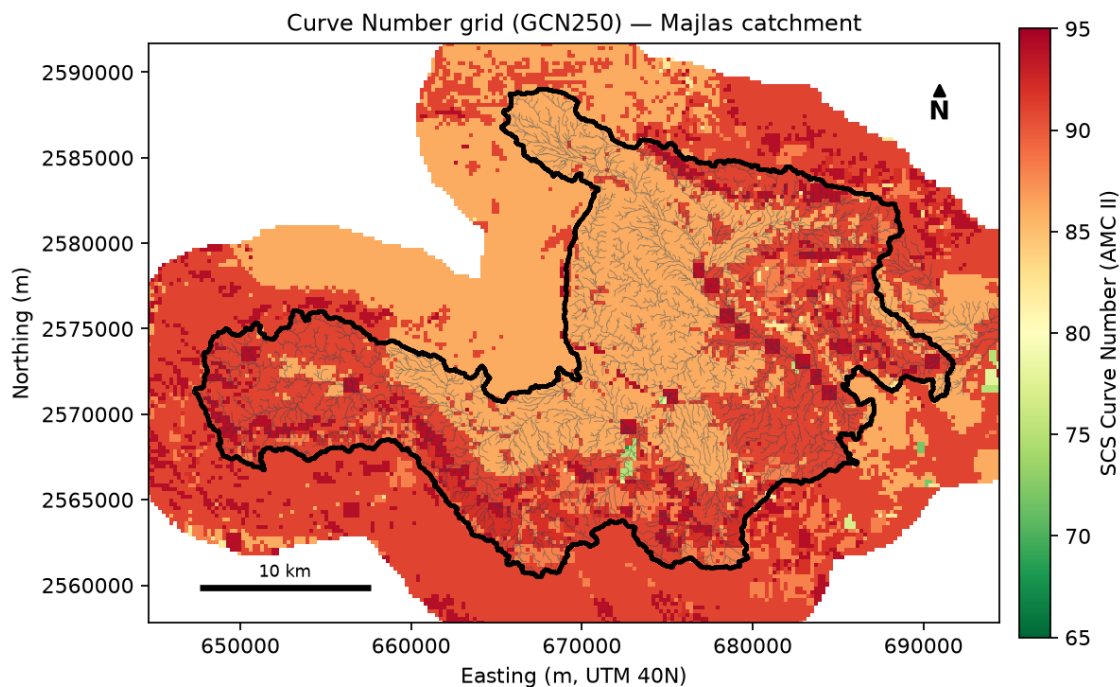


Figure 8 SCS curve-number grid (GCN250) over the Wadi Majlas catchment.

7.3. AREA REDUCTION FACTOR

Point design rainfall over-estimates the average depth over a large catchment. An Area Reduction Factor (ARF), derived from the extreme Oman storms, was applied through the depth-area analysis in the final HEC-HMS model. The ARF used follows:

$$ARF = 1 - 0.001 A^{(1.25 - 0.062 \ln A)}$$

Table 5 Area reduction factor applied in the depth-area analysis.

Area (km2)	Area reduction factor
1	1.000

Area (km2)	Area reduction factor
5	0.994
10	0.987
20	0.976
30	0.965
40	0.956
50	0.948
100	0.913
200	0.864
500	0.776
1000	0.694
2000	0.606

7.4. CATCHMENT AND STREAM DELINEATION

Catchment and stream delineation was performed on the 5 m DTM after sink filling, flow-direction and flow-accumulation processing, using a 1 km² minimum drainage-area threshold. The delineated network reproduces the natural drainage pattern and drains to the proposed dam and, downstream, along the dyked channel. The sub-basin delineation adopted in the HEC-HMS model is shown in Figure 9.



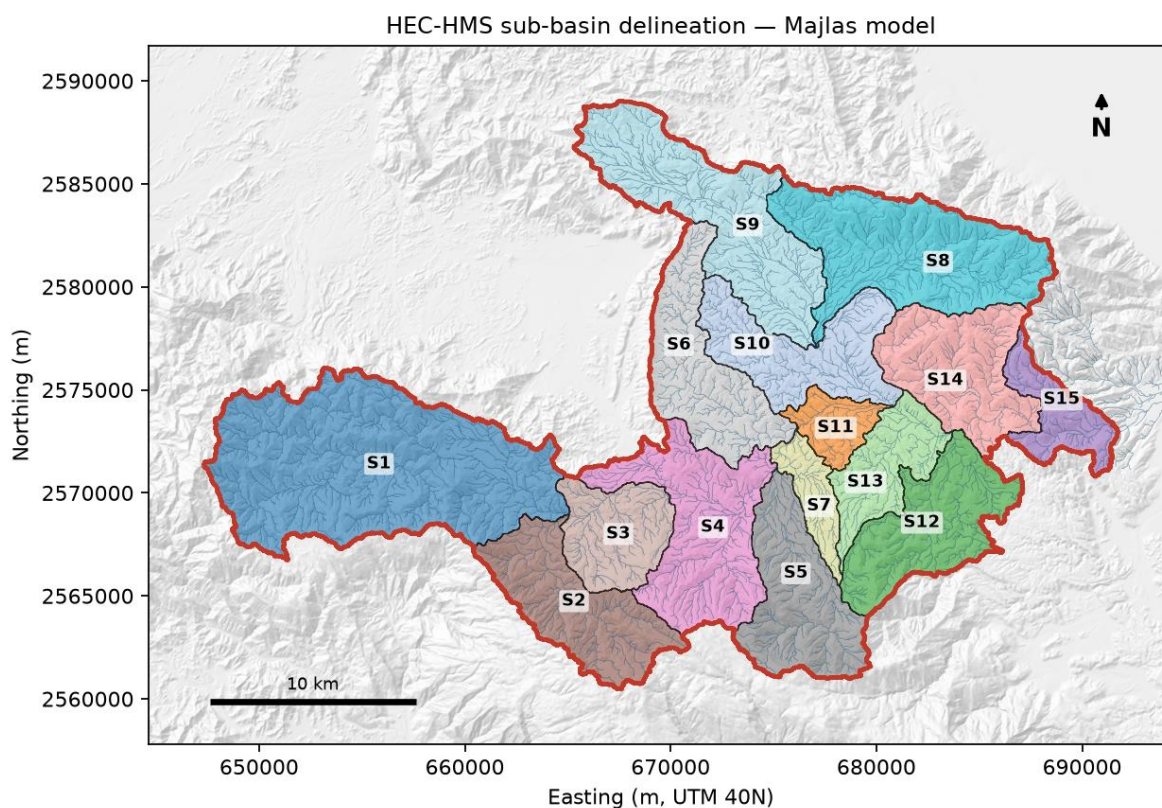


Figure 9 HEC-HMS sub-basin delineation of the Wadi Majlas model.

7.5. SUB-BASIN CHARACTERISTICS AND LAG TIME

Sub-basin lag times were computed from the Kirpich formula (as implemented in HEC-HMS), and initial abstractions from the curve number:

$$t_{lag} = 0.6 \frac{1}{52} \frac{(L_{10-85} \times 1000)^{0.77}}{S^{0.385}}$$

where $L_{(10-85)}$ is the 10-85% flow-path length (km) and S the corresponding slope. The characteristics of the 20 model sub-basins - the 15 upstream sub-basins draining to the dam and

the additional downstream sub-basins introduced for the dyked-channel design - are summarised in Table 8.

Table 6 Characteristics of the HEC-HMS sub-basins (Wadi Majlas model).

Sub-basin	Area (km2)	CN	Ia (mm)	Lag (min)	Downstream
S1	106.8	90.3	5.45	110	J1
S2	36.6	91	5.04	34.8	J1
S3	24.2	87.7	7.12	44	J2
S4	47.9	88	6.94	80.4	J3
S5	36.8	90	5.63	46.9	J3
S6	36.7	86.5	7.94	113	J4
S7	10.7	86.8	7.76	60.8	J4
S9	56.5	86.5	7.92	147	J5
S10	34.2	87.7	7.13	92.3	J6
S11	10.6	87.6	7.21	46.2	J6
S12	37.0	90.5	5.36	65.5	J7
S13	21.6	89.8	5.78	64.6	J7
S14	37.3	90	5.62	70.6	J8
S15	15.9	89.4	6.05	30.3	Majlas Dam
Subbasin-1	40.4	90.1	5.55	43.2	J9
Subbasin-2	18.0	89.2	6.17	43.5	J5
Sdn1	3.0	88	6.92	28.1	J10
Sdn2	20.3	88.5	6.58	24.9	J11
Sdn3	3.5	88.6	6.52	25.7	O1
Sdn4	6.2	88.4	6.65	33.6	O1
TOTAL	604.3				



8. MODEL CALIBRATION AND REVIEW OF THE PREVIOUS STUDY

The catchment model was calibrated against observed hydrographs from Cyclone Gonu (June 2007), the largest instrumented event in the region, and checked against Cyclone Phet (2010). The previous study calibrated the model to observed discharge hydrographs at three cross-sections near the dam site, including the Majlass and Hayfadh gauges.

During the present re-calibration, two of the three observed hydrographs could be reproduced with consistent, physically reasonable parameters. The third - the observed hydrograph at the gauge nearest the Majlas dam - could not be matched with the gauged Gonu rainfall. A sensitivity analysis was carried out in which the recorded storm rainfall at the contributing gauges was progressively reduced to determine the reduction required to reproduce that observed hydrograph. As Table 9 shows, matching it demanded reductions of the gauged 24-hour Gonu rainfall to physically implausible levels at several stations (for example from 384 mm to 38 mm at Buei and from 231 mm to 23 mm at Qurayat), while other stations required almost no change.

Table 7 Rainfall reduction required to reproduce the questioned observed hydrograph (Cyclone Gonu, 2007).

Gauge	Gonu 24-h (mm)	Reduced to match (mm)	Reduction
Jabal Abyadh East	629	252	-60%
Buei near Buei	384	38	-90%
Taba at Taba	518	414	-20%
Hayfadh at Hayfadh	625	625	0%
Qurayat at Qurayat	231	23	-90%

Such non-uniform, extreme reductions have no physical basis. The unavoidable conclusion is that the third **observed hydrograph used in the previous study is erroneous** - most probably affected by a rating or gauging error at high stage - and should not have been used as a calibration target. This inconsistency was not identified or flagged in the previous report. The present model was therefore calibrated to the two reliable observed hydrographs and to observed peak flows, and the questioned record was excluded.



9. DESIGN FLOOD RESULTS

Design floods were computed for return periods from 2 to 10,000 years and for the PMP, using the Flood-Mapping design rainfall and the calibrated HEC-HMS model. Two configurations were run:

- a without-reservoir configuration (Dams runs), representing the catchment with the dam sites present as flow nodes but without reservoir storage (Majlas and Haifaz); and
- a with-reservoir configuration (Res runs), in which the Majlas reservoir storage and spillway are represented and the inflow is routed through the reservoir.

The Majlas reservoir has a full-supply level of 83 m and a crest level of 94 m, providing about 94 Mm³ of storage (dam height about 74.6 m). Its elevation-area-storage relationship, derived from the drone survey, is shown in Figure 10.

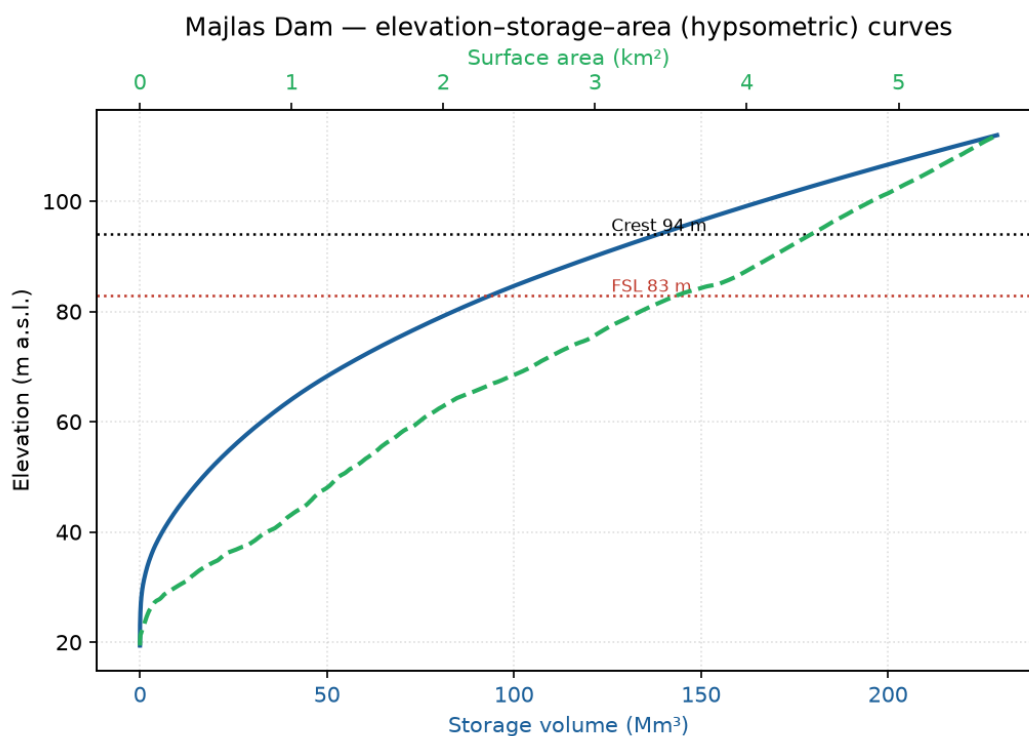


Figure 10 Majlas Dam elevation-storage-area (hypsimetric) curves.

9.1. PEAK DISCHARGES

Table 10 lists peak discharges at key locations for selected return periods, with and without the Majlas reservoir. Figure 11 shows the peak-discharge frequency curve at the catchment outlet for both configurations.



Table 8 Design peak discharges (m³/s) at key locations, with and without the Majlas reservoir.

Location / configuration	2-yr	10-yr	100-yr	1,000-yr	10,000-yr	PMP
Majlas Dam (inflow) - no res.	92	1,161	3,452	8,289	14,166	18,726
Catchment outlet O1 - no res.	86	1,126	3,351	8,026	13,798	18,692
Downstream J10 - no res.	89	1,144	3,398	8,131	13,923	18,532
Downstream J12 - no res.	88	1,135	3,394	8,139	13,968	18,751
Majlas Dam (routed out) - with res.	0	0	0	2,228	8,749	17,224
Catchment outlet O1 - with res.	17	216	449	2,180	8,585	17,286
Downstream J10 - with res.	2	29	56	2,211	8,665	17,123
Downstream J12 - with res.	17	221	430	2,198	8,664	17,303

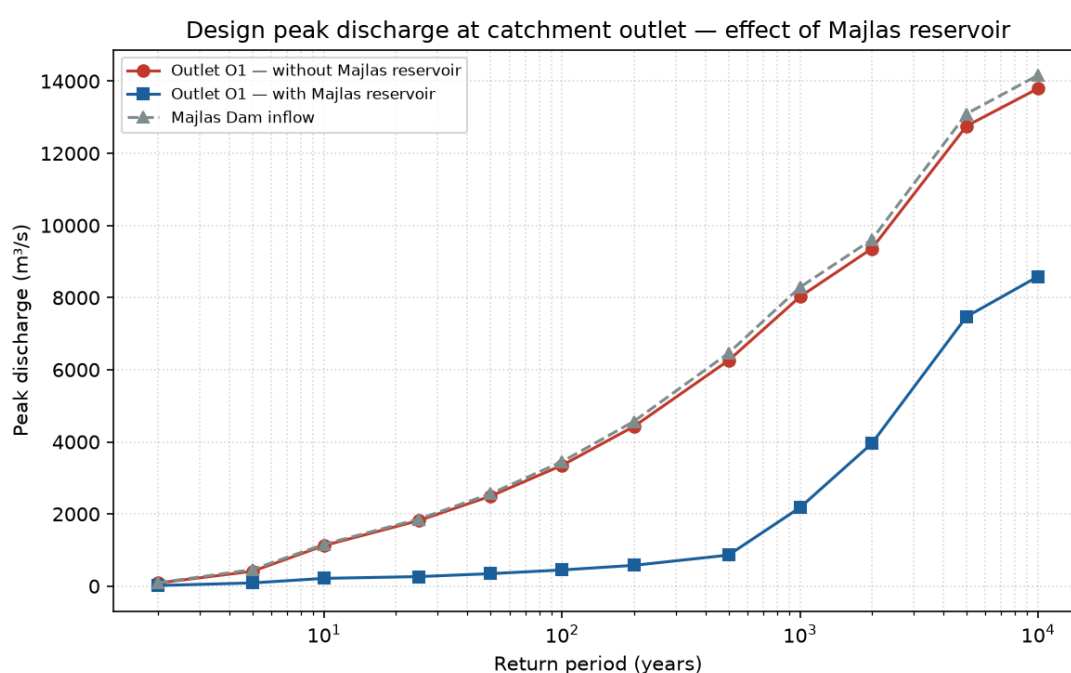


Figure 11 Peak-discharge frequency at the catchment outlet - effect of the Majlas reservoir.

9.2. RESERVOIR ROUTING AND ATTENUATION

The Majlas reservoir provides substantial flood attenuation. It stores the full design inflow up to about the 200-year event (zero spillway outflow), and strongly attenuates larger floods. Table 11 and Figures 12 and 13 summarise the inflow, routed outflow and attenuation. For the 10,000-year event the peak is reduced from 14,166 m³/s to 8,749 m³/s (38% attenuation); even for the PMP the routed outflow (17,224 m³/s) remains below the inflow (18,726 m³/s).



Table 9 Majlas reservoir routing - peak inflow, routed outflow and attenuation.

Return period	Peak inflow (m3/s)	Peak outflow (m3/s)	Attenuation
2-yr	92	0	100%
5-yr	460	0	100%
10-yr	1,161	0	100%
25-yr	1,859	0	100%
50-yr	2,561	0	100%
100-yr	3,452	0	100%
200-yr	4,566	0	100%
500-yr	6,459	877	86%
1000-yr	8,289	2,228	73%
2000-yr	9,607	4,029	58%
5000-yr	13,095	7,608	42%
10000-yr	14,166	8,749	38%
PMP	18,726	17,224	8%

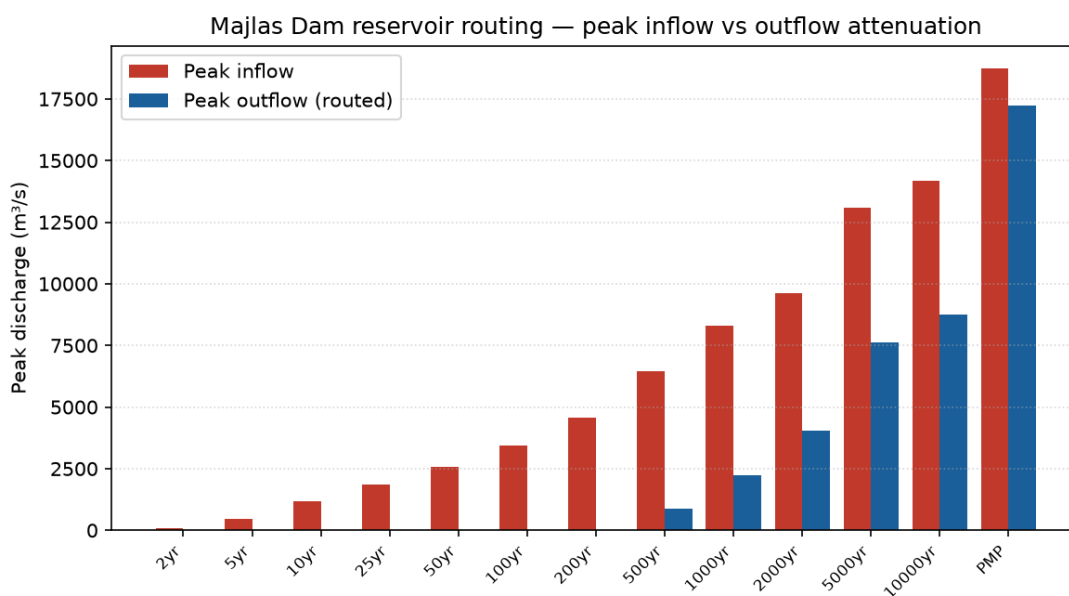


Figure 12 Majlas reservoir routing - peak inflow versus routed outflow by return period.



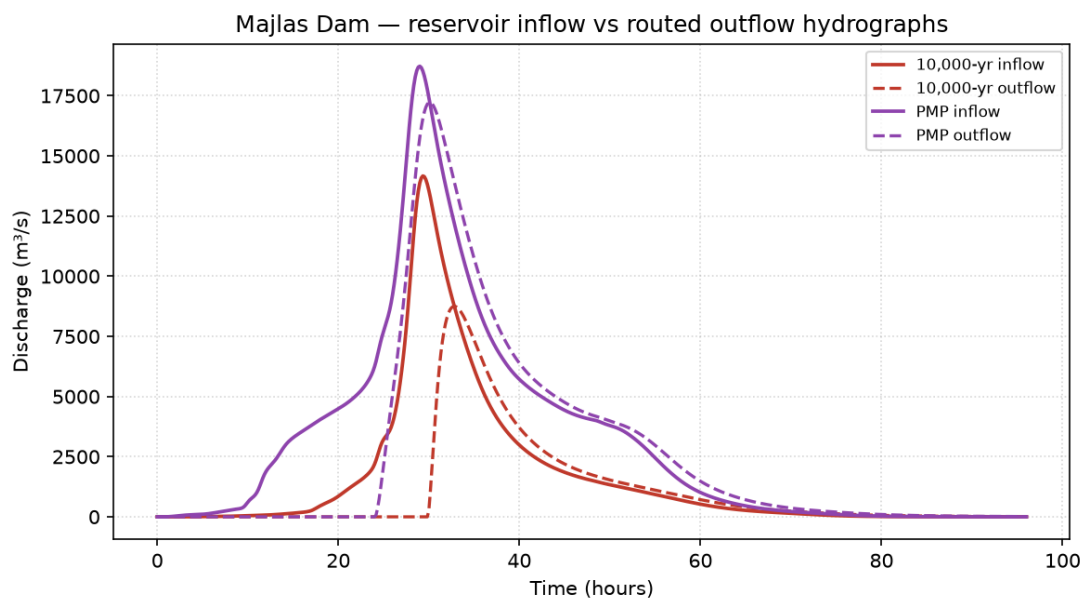


Figure 13 Majlas reservoir inflow and routed-outflow hydrographs (10,000-year and PMP events).

The downstream peak discharges (locations J10, J12 and the outlet O1 in Table 10) provide the design flows for the dyked-channel protection works. Because the reservoir removes the upstream flood contribution up to moderate return periods, the design flows for the downstream channel under the with-reservoir configuration are governed primarily by the local downstream sub-catchments for events up to about the 200-year return period, and by reservoir spill for rarer events.

10. CONCLUSIONS AND RECOMMENDATIONS

The updated hydrological study for the Wadi Majlas Flood Protection Dam supports the following conclusions:

- The catchment draining to the dam is 571 km² (604 km² including the downstream sub-catchments), mountainous and steep, with a catchment-average curve number of about 89.
- Local GEV/L-moment rainfall frequency analysis agrees closely with the Oman Flood-Mapping design rainfall up to the 10-year event but diverges strongly at rare return periods owing to the dominance of Cyclone Gonu (2007) in the short records. The Flood-Mapping depths were adopted for design.
- One of the three observed hydrographs used for calibration in the previous study is erroneous: it can only be reproduced with physically implausible rainfall reductions. It was excluded, and this error was not flagged previously.
- The Majlas reservoir stores the full design inflow up to about the 200-year event and attenuates the 10,000-year peak by about 38%, from 14,166 to 8,749 m³/s.
- Design peak discharges and hydrographs for return periods 2-10,000 years and the PMP are provided for the dam spillway and the downstream dyked-channel design.

It is recommended that the design proceed on the Flood-Mapping-based design floods with reservoir routing, that the questioned historical gauge record be formally set aside, and that the local rainfall analysis be revisited as additional years of record become available.



11. REFERENCES

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